

JEE Main - 1 | JEE - 2023 (Gen 1 & 2)

Date: 08/06/2021

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 9.
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

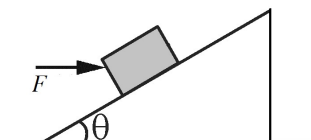
OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

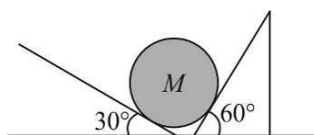
PART - I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

- A hot-air balloon starts rising from the ground with constant acceleration 2.5 m/s^2 . After 10 s, a stone is released from the balloon. Until it reaches the ground, the total distance covered by the stone is:
($g = 10 \text{ m/s}^2$)
(A) 175 m (B) 187.5 m (C) 225 m (D) 237.5 m
- Two cars A and B move on two straight roads, both at speed 40 km/h. If the angle between the velocities of the cars is 60° , the velocity of A relative to B is:
(A) $10\sqrt{5} \text{ km/h}$ (B) $40\sqrt{3} \text{ km/h}$ (C) 40 km/h (D) $\frac{40}{\sqrt{3}} \text{ km/h}$
- When a person is running with velocity $5\hat{i} + 10\hat{j}$, the wind appears to come from the direction $\hat{i}$. When he triples his velocity, the wind appears to come from the direction $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$. Then, the true velocity of the wind is:
(A) $4\hat{i} + 8\hat{j}$ (B) $5\hat{i} + 10\hat{j}$ (C) $-5\hat{i} + 10\hat{j}$ (D) $-4\hat{i} + 8\hat{j}$
- A block of mass m is placed on a fixed inclined plane. If the plane is frictionless, the horizontal force that must be applied on the block to keep it in equilibrium is:

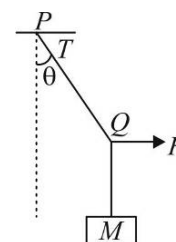


- (A) $mg \sin \theta$ (B) $mg \cos \theta$ (C) $mg \tan \theta$ (D) $mg(\cos \theta - \sin \theta)$
- When a person is at rest, he observes rain falling with velocity of $20\sqrt{3} \text{ m/s}$ at an angle 60° with the vertical. The velocity (in m/s) at which he should run so that rain appears to fall vertically is:
(A) 30 (B) 10 (C) $10\sqrt{3}$ (D) $20\sqrt{3}$
- A sphere of mass 10 kg rests on two fixed inclined planes as shown. Friction is absent. The contact force (in Newton) acting on the sphere due to the inclined plane of angle 60° is:
($g = 10 \text{ m/s}^2$)

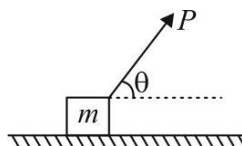


- (A) 25 (B) $50\sqrt{3}$ (C) 50 (D) $25\sqrt{3}$

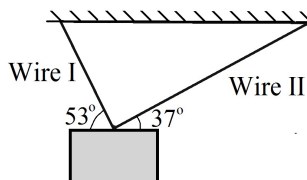
7. A mass of 2.4 kg is suspended by two massless ropes from a rigid support at P with a joint at Q . A constant horizontal force $F = 32$ N is applied at the joint Q . If the block remains in equilibrium in the position shown, the tension in the string PQ (in Newton) is: ($g = 10 \text{ m/s}^2$)



- (A) $20\sqrt{2}$ (B) $16\sqrt{2}$
(C) 32 (D) 40
8. Three forces act on a particle: $\vec{F}_1 = p\hat{i} + \hat{j} - \hat{k}$, $\vec{F}_2 = -2\hat{i} + \hat{j} + 3\hat{k}$ and $\vec{F}_3 = 4\hat{i} + \hat{j} - 2\hat{k}$. If the net force on the particle has magnitude 5 units, the value of p is:
(A) 2 (B) -4 (C) 3 (D) 4
9. Starting from rest, two particles P and Q start moving towards each other with constant acceleration a and $2a$ respectively. If the initial separation between the particles was d , the distance travelled by P until the particles meet is:
(A) $\frac{d}{3}$ (B) $\frac{d}{4}$ (C) $\frac{d}{5}$ (D) $\frac{d}{6}$
10. If $3\hat{i} + 2\hat{j} + \hat{k}$ and $\hat{i} - 2\hat{j} + 3\hat{k}$ represent the adjacent sides of a parallelogram, then the area of this parallelogram is:
(A) $4\sqrt{3}$ (B) $6\sqrt{3}$ (C) $8\sqrt{3}$ (D) $12\sqrt{3}$
11. A force P is applied on a block of mass m placed on a horizontal floor with friction coefficient μ at an angle $\theta = 45^\circ$ with the horizontal as shown. The minimum value of P required to move the block is:

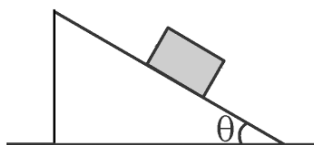


- (A) $\left(\frac{\mu}{\sqrt{2}(\mu-1)}\right)mg$ (B) $\left(\frac{\mu}{\sqrt{2}(\mu+1)}\right)mg$
(C) $\left(\frac{\sqrt{2}\mu}{\mu-1}\right)mg$ (D) $\left(\frac{\sqrt{2}\mu}{\mu+1}\right)mg$
12. A block is suspended using two wires I and II as shown. The maximum tension that the wires can bear without breaking is 300 N and 200 N respectively. The maximum weight of the block (in Newton) such that none of the wires breaks is: ($\cos 37^\circ = 0.8$, $\cos 53^\circ = 0.6$)

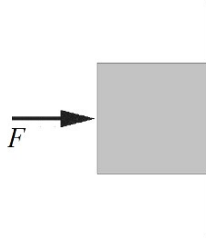


- (A) 300 (B) $\frac{950}{3}$ (C) $\frac{1000}{3}$ (D) 350

13. A particle moving in a straight line covers half the distance with speed 10 m/s. The other half of the distance is covered in two equal time intervals with speed 15 m/s and 5 m/s respectively. The average speed of the particle during the complete motion is:
 (A) 8.5 m/s (B) 10 m/s (C) 12 m/s (D) 10.5 m/s
14. Starting from rest from a point A, a car moves with constant acceleration in a straight line. B and C are two points in its path a distance 168 m apart. The car takes 6 seconds to go from B to C and reaches C with speed 40 m/s. The distance AB is:
 (A) 22.5 m (B) 36 m (C) 24 m (D) 32 m
15. A man is driving a boat in a river with a velocity of $5\hat{i} + 10\hat{j}$ relative to the river water. The river water is flowing with velocity $5\hat{j}$. He observes a helicopter moving with velocity $25\hat{i} + 15\hat{j} - 10\hat{k}$. What is the actual velocity of the helicopter?
 (A) $35\hat{i} + 45\hat{j} - 10\hat{k}$ (B) $30\hat{i} + 35\hat{j} - 10\hat{k}$
 (C) $30\hat{i} + 30\hat{j} - 10\hat{k}$ (D) $30\hat{i} + 45\hat{j} - 10\hat{k}$
16. A block of mass m is placed on a fixed, rough inclined plane. If the block remains in equilibrium, the net force exerted by the plane on the block is:



- (A) $mg \cos \theta$ (B) $mg \sin \theta$ (C) $mg \tan \theta$ (D) mg
17. If $\vec{a} = -\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j} + 2\hat{k}$, the component of $\vec{b}$ along the direction of $\vec{a}$ has magnitude:
 (A) $\frac{5}{3}$ (B) $\frac{2}{3}$ (C) $\frac{4}{9}$ (D) $\frac{5}{9}$
18. A person drops a stone from the top of a tower. If the stone reaches the ground with velocity 18 m/s, the height of the tower is: ($g = 10 \text{ m/s}^2$)
 (A) 14.4 m (B) 16.2 m (C) 19.8 m (D) 21.6 m
19. A person is trying to cross a river in which the water is flowing at 4 km/h. If he can swim at 8 km/hr in still water, at what angle with the direction of flow of the river should he orient his swimming effort if he wants to reach a point on the other bank, directly opposite his starting point?
 (A) 150° (B) 135° (C) 120° (D) 75°
20. A block of weight W is held in equilibrium in contact with a vertical wall by exerting a horizontal force F as shown. If the coefficient of friction between the block and the wall is μ , it can be concluded that:



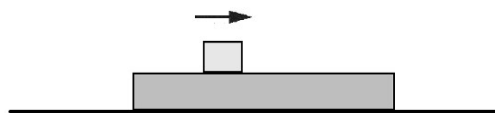
- (A) $\mu \geq \frac{F}{W}$ (B) $\mu = \frac{F}{W}$ (C) $\mu \geq \frac{W}{F}$ (D) $\mu = \frac{W}{F}$

SPACE FOR ROUGH WORK

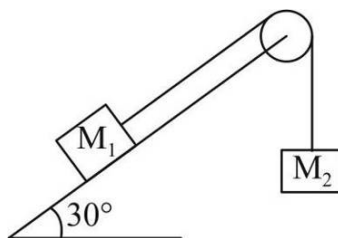
SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9.

21. A plank of mass 3 kg remains stationary on the horizontal surface when a block of mass 1 kg moves on it as shown. If the coefficient of friction between the block and the plank is 0.2 and the coefficient of friction between the plank and the surface is 0.1, the force of friction (in Newton) exerted by the surface on the plank is _____. ($g = 10 \text{ m/s}^2$)



22. Starting from rest, a particle moves in a straight line with uniform acceleration 1 m/s^2 . The displacement (in metres) of the particle in the third second of its motion is _____.
23. Starting from rest, a car accelerates uniformly for 2 seconds, then moves at constant velocity 12 m/s for 3 seconds, and then decelerates uniformly for 3 seconds until it comes to rest. If the total distance (in metres) covered by the car is $10X$, the value of X is _____.
24. In the figure shown, the inclined plane is fixed and frictionless, the string is massless and inextensible, and the pulley is light and frictionless. If $M_1 = 6 \text{ kg}$, find M_2 (in kg) required to keep the system in equilibrium.



25. Two girls are running on a railway platform in opposite directions, both parallel to the railway track at constant speed 3 m/s each. A train moving at constant speed runs over the track beside them. The time taken by the train to cross one girl is 10 s and to cross the other girl is 15 s . If the length of the train (in metres) is $20X$, the value of X is _____.

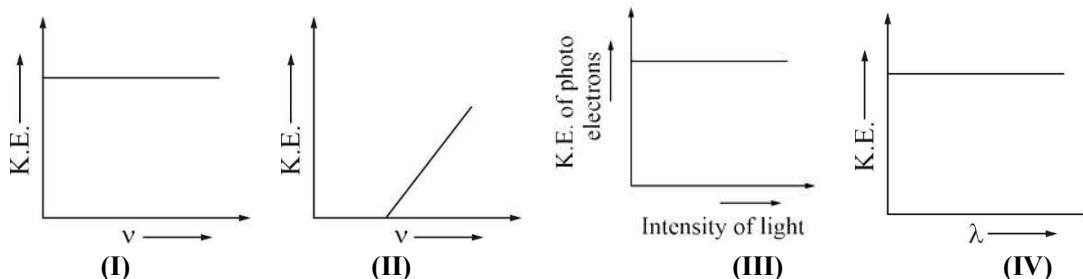
SPACE FOR ROUGH WORK

PART - II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- Find the weight of H_2SO_4 in 1200 mL of 0.2 N H_2SO_4 solution:
(M_0 weight of H = 1u, S = 32u, O = 16u)
(A) 23.52 g (B) 2.52 g (C) 3.52 g (D) 11.76 g
- 10g of MgCO_3 decomposes on heating to 0.1 mole CO_2 and 4 g MgO . The percent purity of MgCO_3 is : (Given that atomic weights of Mg, C and O are 24, 12 and 16u)
(A) 44% (B) 54% (C) 74% (D) 84%
- Assume 60% w/w aqueous solution of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) and 20% w/w aqueous solution of urea (NH_2CONH_2) having equal molarity, then which solution has higher density :
(M_0 of C = 12, H = 1, O = 16)
(A) Both have equal density (B) Glucose solution
(C) Urea solution (D) Cannot be predicted
- Percentage loss in mass, when $\text{NaHCO}_3(\text{s})$ is heated in open vessel. (M_0 of $\text{NaHCO}_3 = 84\text{g}$)
(A) 21.12% (B) 36.9% (C) 30% (D) 32.23%
- Which hydrogen like species will have same radius as that of first Bohr's orbit of hydrogen atom ?
(A) $n = 2, \text{Be}^{3+}$ (B) $n = 2, \text{Li}^{2+}$ (C) $n = 2, \text{He}^+$ (D) $n = 3, \text{Li}^{2+}$
- The ratio of minimum to maximum wavelengths in Balmer series of H-atom is :
(A) 5 : 9 (B) 5 : 36 (C) 1 : 4 (D) 3 : 4
- The angular momentum of electron in an orbit of Li^{2+} is found to be $14\left(\frac{h}{11}\right)$. Calculate the potential energy (in eV) of system. $\left(\pi = \frac{22}{7}\right)$
(A) $13.6 \times \left(\frac{3}{8}\right)^2$ (B) $-13.6 \times \left(\frac{3}{8}\right)^2$
(C) $-2 \times 13.6 \times \left(\frac{3}{8}\right)^2$ (D) $-2 \times 13.6 \times \left(\frac{8}{3}\right)^2$
- Amongst the following statements, which statements was proposed by Dalton.
(A) All the atoms of a given element have identical properties including mass. Atoms of different elements differ in mass.
(B) Chemical reactions involve reorganization of atoms. These are neither created nor destroyed in a chemical reaction
(C) Matter consists of indivisible atoms
(D) All of these

9. A mixture of NaOH and Na_2CO_3 required 25 mL of 0.1 M HCl using phenolphthalein as the indicator. However, the same amount of the mixture required 30 mL of 0.1 M HCl when methyl orange was used as the indicator. The molar ratio of NaOH and Na_2CO_3 in the mixture was :
 (A) 2 : 1 (B) 1 : 2 (C) 4 : 1 (D) 1 : 4
10. 1 gram of a sample of CaCO_3 was strongly heated and the CO_2 liberated was absorbed in 100 mL of 0.5 M NaOH solution. Assuming 90% purity for the sample, how many mL of 0.5 M HCl would be required to react with the resulting solution to reach the end point in presence of phenolphthalein? (M_0 of $\text{CaCO}_3 = 100\text{g}$)
 (A) 73 mL (B) 41 mL (C) 82 mL (D) 100 mL
11. What is the ratio of time periods (T_1 / T_2) in second orbit of hydrogen atom to third orbit of He^+ ion?
 (A) 8/27 (B) 32/27 (C) 27/32 (D) None of these
12. Which is/are correct graph?



- (A) I, III and IV (B) II and III (C) II, III and IV (D) Only II
13. $n = 3$
 $n = 2$
 $n = 1$
 For the above transitions in hydrogen like atoms, select the correct relation:
 I. $v_c = v_a + v_b$ II. $v_c = \frac{v_a v_b}{v_a + v_b}$ III. $\lambda_c = \lambda_a + \lambda_b$ IV. $\lambda_c = \frac{\lambda_a \lambda_b}{\lambda_a + \lambda_b}$
 (A) Only II, III (B) Only I (C) Only I, IV (D) Only II, IV
14. Hydrogen atoms are excited to $n = 4$ state. In the spectrum of emitted radiation, the ratio of number of lines in the ultraviolet and visible regions are respectively:
 (A) 3 : 1 (B) 1 : 3 (C) 2 : 3 (D) 3 : 2
15. The following process has been used to obtain iodine from oil fields in California :
 $\text{NaI} + \text{AgNO}_3 \longrightarrow \text{AgI} + \text{NaNO}_3$
 $2\text{AgI} + \text{Fe} \longrightarrow \text{FeI}_2 + 2\text{Ag}$
 $2\text{FeI}_2 + 3\text{Cl}_2 \longrightarrow 2\text{FeCl}_3 + 2\text{I}_2$
 How many kg of AgNO_3 is required in the first step for every kg of I_2 produced in the third step ? (M_0 of $\text{AgNO}_3 = 170\text{g}$, $\text{AgI} = 235\text{g}$, $\text{FeI}_2 = 310\text{g}$, $\text{I}_2 = 254\text{g}$)
 (A) 2.68 kg (B) 1.92 kg (C) 0.66 kg (D) 1.34 kg

16. When 80 ml of 0.2 M HNO_3 is added to 120 ml of 0.150 M KOH , the reaction
- $$\text{HNO}_3 + \text{KOH} \longrightarrow \text{KNO}_3 + \text{H}_2\text{O}$$
- Occurs and the resulting solution is :
- (A) $[\text{K}^+] = 0.01, [\text{NO}_3^-] = 0.08$ (B) $[\text{K}^+] = 0.09, [\text{NO}_3^-] = 0.08$
 (C) $[\text{K}^+] = 0.08, [\text{NO}_3^-] = 0.04$ (D) $[\text{K}^+] = 0.08, [\text{NO}_3^-] = 0.08$
17. The ratio of the energy of the electron in ground state of hydrogen to the energy of electron in first excited state of Be^{3+} is :
- (A) 1 : 4 (B) 1 : 8 (C) 1 : 16 (D) 16 : 1
18. The triad of nuclei that is isotonic is :
- (A) ${}^{14}_6\text{C}, {}^{15}_7\text{N}, {}^{17}_9\text{F}$ (B) ${}^{12}_6\text{C}, {}^{14}_7\text{N}, {}^{19}_9\text{F}$ (C) ${}^{14}_6\text{C}, {}^{14}_7\text{N}, {}^{17}_9\text{F}$ (D) ${}^{14}_6\text{C}, {}^{14}_7\text{N}, {}^{19}_9\text{F}$
19. The ratio of kinetic energy and total energy of an electron in a Bohr orbit of a hydrogen-like species is :
- (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 1 (D) -1
20. 0.1 g of a solution containing Na_2CO_3 and NaHCO_3 requires 10 mL of 0.01 N HCl for neutralization using phenolphthalein as an indicator. Mass % of Na_2CO_3 in solution is :
- (A) 25 (B) 32 (C) 50 (D) None of these

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9.

21. The ratio of the masses of methane and ethane in a gas mixture is 4 : 5. The ratio of number of their molecules in the mixture is x : y. The value of x + y is : (M_0 of C = 12, H = 1)
22. 100 ml solution of 0.1 N H_2SO_4 was titrated with 0.1 M KOH solution. The titration was discontinued after adding 40 ml KOH solution. The remaining titration was completed by adding 0.5 M $\text{Ca}(\text{OH})_2$ solution. The volume of $\text{Ca}(\text{OH})_2$ required for completing the titration is :
23. The total number of moles of neutrons present in 108 mL of $\text{H}_2\text{O}(\ell)$ is 6x. Value of x is : (Density of $\text{H}_2\text{O} = 1 \text{ g / mL}$)
24. 2.0 g of polybasic organic acid (Molecular mass = 600) required 100 mL of a $\frac{M}{6}$ NaOH solution for complete neutralisation. Find the basicity of acid.
25. 245 g impure sample of $\text{KClO}_3(\text{s})$ on heating gives 8.64 g of O_2 according to the reaction $2\text{KClO}_3(\text{s}) \longrightarrow 2\text{KCl}(\text{s}) + 3\text{O}_2(\text{g})$. Calculate % purity of sample ? (Molar mass of $\text{KClO}_3 = 122.5 \text{ g}$)

SPACE FOR ROUGH WORK

PART - III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- If the equation, $x^2 + bx + 45 = 0$ ($b \in R$) has conjugate complex roots and they satisfy $|z + 1| = 2\sqrt{10}$, then :
 (A) $b^2 - b = 30$ (B) $b^2 + b = 12$ (C) $b^2 + b = 72$ (D) $b^2 - b = 42$
- If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}$ ($i = \sqrt{-1}$), then $(1 + iz + z^5 + iz^8)^9$ is equal to :
 (A) $(-1 + 2i)^9$ (B) 0 (C) -1 (D) 1
- If $z_1 = \sqrt{3} + i\sqrt{3}$ and $z_2 = \sqrt{3} + i$, then in which quadrant $\left(\frac{z_1}{z_2}\right)$ lies?
 (A) I (B) II (C) III (D) IV
- The root of the equation $2(1+i)x^2 - 4(2-i)x - 5 - 3i = 0$ which has greater modulus is :
 (A) $\frac{3-5i}{2}$ (B) $\frac{5-3i}{2}$ (C) $\frac{3-i}{2}$ (D) None of these
- Value of $\frac{(\cos \theta + i \sin \theta)^4}{(\cos \theta - i \sin \theta)^3}$ is :
 (A) $\cos 5\theta + i \sin 5\theta$ (B) $\cos 7\theta + i \sin 7\theta$
 (C) $\cos 4\theta + i \sin 4\theta$ (D) $\cos \theta + i \sin \theta$
- The complex number z satisfying the equation $|z| - 4 = 0, |z - i| = |z + 5i|$ is :
 (A) $\sqrt{3} - i$ (B) $2\sqrt{3} - 2i$ (C) $-2\sqrt{3} + 2i$ (D) 0
- If $|a + ib| = 1$, then the simplified form of $\frac{1+b+ai}{1+b-ai}$ is :
 (A) $b + ai$ (B) $a + bi$ (C) $(1+b)^2 + a^2$ (D) ai
- If ω is a non-real cube root of unity, then $\frac{1+2\omega+3\omega^2}{2+3\omega+\omega^2} + \frac{2+3\omega+3\omega^2}{3+3\omega+2\omega^2}$ is equal to :
 (A) -2ω (B) 2ω (C) ω (D) 0
- If $z = 2 + i$, then $(z-1)(\bar{z}-5) + (\bar{z}-1)(z-5)$ is equal to :
 (A) 2 (B) 7 (C) -1 (D) -4
- If a complex number z satisfies the equation $z + \sqrt{2}|z+1| + i = 0$, then $|z|$ is equal to :
 (A) 2 (B) $\sqrt{3}$ (C) $\sqrt{5}$ (D) 1
- Number of solutions of the equation $|z|^2 + 7\bar{z} = 0$ is/are :
 (A) 1 (B) 2 (C) 4 (D) 6
- If $1, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$ are the n^{th} roots of unity, then $(2 - \alpha_1)(2 - \alpha_2) \dots (2 - \alpha_{n-1})$ is equal to :
 (A) n (B) 2^n (C) $2^n + 1$ (D) $2^n - 1$

13. Let $\omega = e^{i2\pi/3}$ and a, b, c, x, y, z be non-zero complex numbers such that
 $a + b + c = x; a + b\omega + c\omega^2 = y; a + b\omega^2 + c\omega = z$
 Then the value of $\frac{|x|^2 + |y|^2 + |z|^2}{|a|^2 + |b|^2 + |c|^2}$ is :
 (A) 1 (B) 2 (C) 3 (D) 4
14. Number of roots of the equation $|2x-1|^2 - 3|2x-1| + 2 = 0$ is:
 (A) 1 (B) 2 (C) 3 (D) 4
15. If the roots of $(a^2 + b^2)x^2 - 2(bc + ad)x + c^2 + d^2 = 0$ are equal then :
 (A) $\frac{a}{b} = \frac{c}{d}$ (B) $\frac{a}{c} + \frac{b}{d} = 0$ (C) $\frac{a}{d} = \frac{b}{c}$ (D) $a + b = c + d$
16. The value of a for which the sum of the squares of the roots of the equation $x^2 - (a-2)x - a - 1 = 0$ assume the least value is :
 (A) 2 (B) 3 (C) 0 (D) 1
17. If α and β be the roots of $x^2 + px + q = 0$, then $\frac{(\omega\alpha + \omega^2\beta)(\omega^2\alpha + \omega\beta)}{\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}}$ (ω is non real cube root of unity) is equal to :
 (A) $-\frac{q}{p}$ (B) $\alpha\beta$ (C) $-\frac{p}{q}$ (D) ω
18. If α and β are the roots of $x^2 + x + 2 = 0$ and γ, δ roots of $x^2 + 3x + 4 = 0$ then $(\alpha + \gamma)(\alpha + \delta)(\beta + \gamma)(\beta + \delta)$ is equal to :
 (A) -18 (B) 18 (C) 24 (D) 44
19. If λ is an integer and α, β are the roots of $4x^2 - 16x + \frac{\lambda}{4} = 0$ such that $1 < \alpha < 2$ and $2 < \beta < 3$, then number of possible values of λ is:
 (A) 3 (B) 17 (C) 15 (D) None of these
20. If both the roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then k lies in the interval:
 (A) $[4, 5]$ (B) $(-\infty, 4)$ (C) $(6, \infty)$ (D) $(5, 6)$

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9.

21. If α and β be the values of x in $m^2(x^2 - x) + 2mx + 3 = 0$ and m_1 and m_2 be two values of m for which α and β are connected by the relation $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{4}{3}$. Then the value of $\left| \frac{m_1^2}{m_2} + \frac{m_2^2}{m_1} \right|$ can be expressed as $\frac{p}{q}$, where p and q are coprime. Then sum of digits of $p + q$ is _____.
22. For a complex number z , the minimum value of $|z| + |z - 2|$ is _____.
23. If x is real, the maximum value of $\frac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$ is M , then $M - 36$ is _____.
24. If α and β are the roots of equation $x^2 - 2x + 4 = 0$ then the value of $\frac{\alpha^6 + \beta^6}{(\alpha\beta)^2}$ will be:
25. The possible greatest integral value of a for which the expression $\frac{ax^2 + 3x + 4}{x^2 + 2x + 2}$ is less than 5 for all real x is _____.

SPACE FOR ROUGH WORK

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